

GAP ANALYSIS

PART REQUIREMENTS FOR SEMINAR

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AOI: INTELLIGENT PRODUCT TRACEABILITY

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OVERVIEW OF IDENTIFIED RESEARCH GAPS

Intelligent Product Traceability — Five Thematic Research Gaps

THEME	GAP TITLE	KEY SUB-VARIABLES
1	Intelligent Product Traceability and Blockchain Technology	Smart Contracts, Decentralisation, Scalability, Data Integrity, Interoperability
2	Intelligent Product Traceability and IoT Integration	Real-Time Sensing, RFID/QR Tracking, Edge Computing, Data Security, Connectivity
3	Intelligent Product Traceability and Supply Chain Transparency	Provenance Verification, Information Sharing, Stakeholder Collaboration, Visibility, Fraud Reduction
4	Intelligent Product Traceability and Consumer Trust / Food Safety	Traceability Label Trust, Purchase Behaviour, Technology Acceptance, Safety Verification, Willingness to Pay
5	Intelligent Product Traceability and Organisational Performance / Sustainability	Operational Efficiency, Sustainability, Counterfeit Reduction, Supply Chain Integrity, Competitive Advantage

GAP ONE

**INTELLIGENT PRODUCT TRACEABILITY
AND BLOCKCHAIN TECHNOLOGY**

THEME 1:

INTELLIGENT PRODUCT TRACEABILITY

(Its Sub-Variables) and Blockchain Technology

x1

Smart Contracts

x2

Decentralisation

x3

Scalability

x4

Data Integrity

x5

Interoperability

Gap 1: Intelligent Product Traceability and Blockchain Technology

THEME 1: INTELLIGENT PRODUCT TRACEABILITY

GAP 1: INTELLIGENT PRODUCT TRACEABILITY AND BLOCKCHAIN TECHNOLOGY

Blockchain technology has emerged as one of the most transformative infrastructures underpinning intelligent product traceability systems across multiple industries. Its distributed, immutable ledger design addresses the fundamental limitations of centralized traceability databases, which are inherently vulnerable to single-point failures and data manipulation. Wong, Ting and Atanda (2024) demonstrate that the convergence of blockchain with the Internet of Things and machine learning has the potential to fully transform supply chain traceability by enabling end-to-end visibility and secure real-time data sharing. Similarly, Hasan *et al.*, (2023) developed OriginChain, a peer-to-peer blockchain architecture that replaces centralized databases with distributed ledgers, providing tamper-proof traceability records for agricultural products and automating regulatory compliance checks across supply chain participants.

Despite these notable advances, a critical and persistent gap surrounds the issue of scalability. As Cordeiro and Ferreira (2025) observe, current blockchain systems struggle significantly with transaction volume and processing speed in large-scale supply chains, creating serious operational bottlenecks when thousands of nodes simultaneously attempt to write or query traceability records. Lv *et al.*, (2023) reinforce this point by highlighting that blockchain size grows continuously with each new transaction, since every participating node must maintain a full replica of the entire chain, creating an unsustainable storage burden at industrial scale. The architectural tension between on-chain immutability and practical data storage requirements remains unresolved in the literature, with most proposed solutions relying on hybrid off-chain storage mechanisms such as IPFS that introduce their own access-control vulnerabilities.

Smart contract design represents another deeply under-explored dimension of blockchain-based traceability. Zhang and Cheng (2023) identify that the structural complexity of smart contracts can dramatically increase transaction costs when deployed on public blockchains, yet most studies treat smart contracts as a given feature rather than a design object requiring quality engineering. Yao and Zhang (2022) further reveal that even in well-designed systems, the authenticity of data uploaded by enterprises cannot be fully guaranteed since blockchain immutability simply locks in whatever information is entered at the point of origin, creating a fundamental 'garbage in, garbage out' problem. Viji *et al.*, (2022) extend this concern to governance, noting that blockchain technology still faces unresolved challenges in identity registration, privacy regulation, and cross-platform standards.

Interoperability across heterogeneous blockchain platforms remains one of the most significant structural gaps in the field. Patel *et al.*, (2023) note that multiple non-standardised blockchain platforms continue to hinder cross-chain collaboration, making it practically impossible for supply chain partners operating on different platforms to share traceability data seamlessly. This fragmentation is especially problematic in global agri-food chains where participants span dozens of countries with different legal and technological environments. Afrin and Pathak (2023) acknowledge that while blockchain's role in shaping the future of global trade is undoubted, the practical maturation of cross-chain interoperability frameworks remains an urgent unmet need. Collectively, these gaps call for future research that addresses scalability, smart contract quality engineering, data authenticity at source, and cross-platform interoperability as integrated design challenges rather than isolated technical problems.

In summary, while blockchain technology offers a compelling architectural foundation for intelligent product traceability, significant gaps persist in its real-world scalability, governance standards, data source authenticity, and interoperability with other systems. Future research must develop robust, tested frameworks that address these gaps holistically, with particular emphasis on large-scale, multi-stakeholder environments in emerging economies where infrastructure limitations compound these technical challenges.

THEME 1: Intelligent Product Traceability and Blockchain Technology

S/N	AUTHOR/DATE	TITLE	GAP STATEMENT
1	Wong et al. (2024)	<i>Enhancing supply chain traceability through blockchain and IoT integration</i>	Additional research and a more extensive examination of the literature is needed; challenges related to security, standardisation, and interoperability must be addressed.
3	Hasan et al. (2023)	<i>A peer-to-peer blockchain-based architecture for trusted agricultural product traceability</i>	Research on blockchain-based systems rarely discusses adaptability. Adopting blockchain is always a challenge to traditional industries due to the learning curve and integration cost.
5	Lv et al. (2023)	<i>Blockchain-based traceability for agricultural products: A systematic literature review</i>	Blockchain size is a design concern because it grows with every transaction and every participant holds a full replica of the chain.
6	Zhang & Cheng (2023)	<i>AI-enabled Product Authentication and Traceability in Global Supply Chains</i>	The smart contract's structural design has a large cost impact on public blockchains. Adaptability is a quality attribute required by many industrial projects that is still unaddressed.
12	Viji et al. (2022)	<i>Blockchain based traceability in supply chain using smart contracts</i>	Blockchain still faces major challenges in scalability, governance, identity registration, privacy, standards and regulation.
33	Yao & Zhang (2022)	<i>Improving agricultural product traceability using blockchain</i>	How to ensure the authenticity and reliability of data at the source is a challenging problem. More large-scale real-world pilot projects are needed to verify robustness.

GAP TWO

**INTELLIGENT PRODUCT TRACEABILITY
AND IoT INTEGRATION**



THEME 2:

INTELLIGENT PRODUCT TRACEABILITY

(Its Sub-Variables) and IoT Integration

x1

Real-Time Sensing

x2

RFID / QR Tracking

x3

Edge Computing

x4

Data Security

x5

Connectivity

Gap 2: Intelligent Product Traceability and IoT Integration

THEME 2: INTELLIGENT PRODUCT TRACEABILITY

GAP 2: INTELLIGENT PRODUCT TRACEABILITY AND IOT INTEGRATION

The Internet of Things has fundamentally expanded the operational scope of intelligent product traceability by enabling the continuous, automated capture of physical-world data across every node of the supply chain. Where traditional traceability systems relied on manual data entry that was inherently slow, error-prone, and susceptible to deliberate falsification, IoT devices including RFID readers, QR code scanners, GPS modules, temperature and humidity sensors now provide a continuous digital thread linking the physical journey of a product to its corresponding data trail. Dib *et al.*, (2023) demonstrate through the Live Data Transmission System model that IoT integration enables real-time information retrieval without errors and allows food companies to detect product anomalies during transportation before they reach consumers, dramatically reducing both waste and safety risks.

Tagarakis *et al.*, (2021) specifically address the persistent gaps in IoT-based traceability systems for fresh produce, noting that despite considerable technical progress, existing implementations remain insufficiently integrated to provide the real-time, end-to-end connectivity required by modern food supply chains. Their work reveals that most current systems operate in isolated data silos where sensor data captured at one node is not automatically accessible to downstream actors, negating the chain-wide visibility that IoT theoretically enables. Xiao *et al.*, (2024) similarly highlight that building a truly functional food traceability system requires clearly defining and linking all participating entities so that information gathered at each stage is recorded, stored, and operationally verified in a unified manner, a systems-integration challenge that existing research has not fully resolved.

A particularly underexplored dimension concerns the security and tamper-resistance of IoT data streams before they reach the blockchain or central database. Adamashvili, Zhizhilashvili and Tricase (2024) explicitly identify that RFID tags and sensors can be replicated and manipulated, creating a critical vulnerability at the physical-digital interface where counterfeit data can be injected into an otherwise secure traceability system. This 'oracle problem', ensuring that physical reality faithfully corresponds to digital records, is acknowledged by Tsang *et al.*, (2019) as one of the most persistent unsolved challenges in IoT-blockchain traceability architectures. Thakur, Ahuja and Singh (2024) add an economic dimension, noting that the cost of deploying large-scale IoT sensor networks and establishing incentive mechanisms for diverse stakeholders to participate remain practically under-investigated, particularly in emerging-market contexts where infrastructure investment capacity is limited.

The challenge of connectivity in low-resource environments is equally underrepresented in current research. Huang, Xie, Ning and Hu (2021) note that QR code recognition is susceptible to interference from external lighting conditions and environmental factors, producing recognition errors that corrupt traceability records. Frikha *et al.*, (2023) further point out that blockchain-IoT integration requires a stable internet connection to function effectively, a prerequisite that cannot be taken for granted in many agricultural regions of Africa and Southeast Asia where product traceability needs are most acute. This creates a significant research gap around the design of low-connectivity, resilient IoT traceability architectures suitable for resource-constrained production environments.

In conclusion, while IoT integration has significantly advanced the real-time data capture capabilities of intelligent product traceability systems, critical gaps remain in physical-digital data integrity, edge-environment connectivity, stakeholder incentive design, and the security of sensor hardware at the point of data collection. Future research must address these gaps through longitudinal, large-scale field implementations in diverse geographical and industrial contexts beyond the controlled laboratory settings that dominate current literature.

THEME 2: Intelligent Product Traceability and IoT Integration

S/N	AUTHOR/DATE	TITLE	GAP STATEMENT
7	Dib et al. (2023)	<i>Application of the Internet of Things in the Traceability of a Supply Chain</i>	GPS and GSM modules are expensive with high energy consumption. Middleware and tags require data encryption for security-critical applications at extra cost.
8	Tagarakis et al. (2021)	<i>Bridging the gaps in traceability systems for fresh produce supply chains</i>	When problems arise in-situ, the system lacks automated escalation protocols; administrators must be manually notified to apply solutions.
20	Huang et al. (2021)	<i>Design of agricultural product traceability platform based on 5G-IoT and blockchain</i>	QR code image recognition is susceptible to light and external environmental interference, resulting in incorrect recognition.
29	Frikha et al. (2023)	<i>Integrating blockchain and deep learning for intelligent greenhouse control and traceability</i>	A stable internet connection is required for the blockchain network. Future research should explore edge computing integration.
37	Tsang et al. (2019)	<i>Blockchain-driven IoT for food traceability with an integrated consensus mechanism</i>	The 'oracle problem', ensuring physical IoT sensor data has not been physically manipulated, remains an unresolved challenge.
42	Thakur et al. (2024)	<i>IoT-GChain: IoT-assisted secure and tractable grain supply chain framework leveraging blockchain</i>	Economic feasibility including deployment costs for IoT sensors and incentive mechanisms for stakeholders to join the network was not addressed.

GAP THREE

INTELLIGENT PRODUCT TRACEABILITY AND SUPPLY CHAIN TRANSPARENCY



THEME 3:

INTELLIGENT PRODUCT TRACEABILITY

(Its Sub-Variables) and Supply Chain Transparency

x1

**Provenance
Verification**

x2

**Information
Sharing**

x3

**Stakeholder
Collaboration**

x4

Visibility

x5

Fraud Reduction

Gap 3: Intelligent Product Traceability and Supply Chain Transparency

THEME 3: INTELLIGENT PRODUCT TRACEABILITY

GAP 3: INTELLIGENT PRODUCT TRACEABILITY AND SUPPLY CHAIN TRANSPARENCY

Supply chain transparency, broadly defined as the capacity of all supply chain actors to access accurate, timely, and trustworthy information about a product's journey from origin to consumer, has been consistently identified in the literature as the primary value proposition of intelligent product traceability systems. Afrin and Pathak (2023) demonstrate that blockchain's creation of transparent, unforgeable digital records facilitates end-to-end visibility into the movement of goods, fundamentally transforming accountability structures in supply chains that have historically operated under significant information asymmetry. Khan *et al.*, (2022) provide empirical evidence confirming that traceability technology has a positive and significant impact on supply chain transparency, and that supply chain transparency in turn has a positive and significant impact on organisational performance, establishing a clear causal pathway from traceability investment to measurable business outcomes.

However, a significant gap exists in understanding how transparency is actually achieved in practice through the coordinated efforts of multiple, often competing, stakeholders. Garcia-Torres, Rey-Garcia and Saenz (2024) make the important theoretical distinction that traceability and transparency are not synonymous: traceability is a technical enabler that generates the data, while transparency is the social and organisational mechanism that converts that data into accountability and trust. Their findings demonstrate that stakeholder collaboration is a fundamental precursor to both constructs, yet existing traceability research predominantly treats transparency as a technical output of system design rather than as a social achievement requiring deliberate governance design. This is a conceptually important but empirically underexplored gap in the literature.

The role of digital platforms in mediating between traceability data and meaningful supply chain transparency is equally underdeveloped. Khan, Alshahrani and Jacquemod (2023) find that digital platforms alone have no significant direct influence on either robust information quality or effective inventory management, suggesting that transparency requires the additional mediation of organisational practices and human interpretation, finding that challenges the dominant techno-optimist narrative in blockchain traceability literature. Karim and Talukder (2024) reinforce this by demonstrating that IoT-blockchain synergies improve real-time transparency but that their benefits are heavily contingent on the quality of pre-existing organisational data management practices rather than on technology deployment alone.

In the context of manufacturing supply chains in the Global South, Langton, Mafini and Tlale (2026) reveal that in South Africa's manufacturing sector, immutability and provenance practices significantly influence supply chain transparency, but that security as a blockchain practice does not, a finding that implies organisations may be over-investing in security features while under-investing in the governance structures that actually drive transparency. Furthermore, Mohammad and colleagues (2026) note that while blockchain-enabled transparency dramatically reduces fraud and improves supply chain performance, future research must develop more realistic, scalable, and economically viable frameworks that move beyond pilot implementations into full operational deployment.

In summary, while intelligent product traceability clearly contributes to supply chain transparency, critical gaps remain in understanding the social and governance dimensions of how traceability data becomes actionable transparency, the mediating roles of digital platforms and organisational capabilities, and the contextual conditions under which transparency benefits are realised differently across industries and geographical settings.

THEME 3: Intelligent Product Traceability and Supply Chain Transparency

S/N	AUTHOR/DATE	TITLE	GAP STATEMENT
31	Khan et al. (2022)	<i>The impact of technologies of traceability and transparency in supply chains</i>	Cross-sectional design limits causal inference. Future studies should use longitudinal designs and extend beyond single-country, single-industry settings.
43	Dietrich et al. (2023)	<i>Blockchain-based traceability architecture for mapping object-related supply chain events</i>	Globalised supply chains cannot rely on focal companies alone to enforce visibility systems. A holistic multi-party architecture for varied event types is lacking.
46	Garcia-Torres et al. (2024)	<i>Enhancing sustainable supply chains through traceability, transparency and stakeholder collaboration</i>	The role of emerging technologies in facilitating collaboration and transparency was not explicitly measured. Longitudinal designs are needed.
48	Langton et al. (2026)	<i>Applying blockchain to improve supply chain transparency, visibility and traceability</i>	Blockchain should be scalable and interoperable with existing SC systems. Further research is needed on SC transparency and its impact on sustainable development.
49	Mohammad et al. (2026)	<i>Enhancing transparency of blockchain technology in supply chain management</i>	Future research ought to make SCM-BT more realistic, effective, and scalable so that supply chain environments may be more open, resilient, and efficient.
51	Khan et al. (2023)	<i>Digital platforms and supply chain traceability for robust information and effective inventory management</i>	Shortage of empirical evidence on how digital platforms and SCT lead to robust information and effective inventory management through transparency mediation.

GAP FOUR

**INTELLIGENT PRODUCT TRACEABILITY
AND CONSUMER TRUST / FOOD SAFETY**

THEME 4:

INTELLIGENT PRODUCT TRACEABILITY

(Its Sub-Variables) and Consumer Trust / Food Safety

x1

Traceability
Label Trust

x2

Purchase
Behaviour

x3

Technology
Acceptance

x4

Safety
Verification

x5

Willingness to
Pay

Gap 4: Intelligent Product Traceability and Consumer Trust / Food Safety

THEME 4: INTELLIGENT PRODUCT TRACEABILITY

GAP 4: INTELLIGENT PRODUCT TRACEABILITY AND CONSUMER TRUST / FOOD SAFETY

The intersection of intelligent product traceability with consumer trust and food safety represents one of the most socially consequential dimensions of this research domain. The ability of traceability systems to assure consumers about the origin, handling, and authenticity of food products is not merely a commercial consideration but a public health imperative, as evidenced by persistent global food safety scandals involving contaminated produce, mislabelled meat, and counterfeit medicines. Lin *et al.*, (2021) demonstrate that blockchain technology significantly improves food traceability and supply chain operations, yet despite these documented benefits, blockchain implementations in food supply chains continue to attract criticism for lacking a comprehensive adoption roadmap that bridges the gap between technical capability and consumer-facing application.

A foundational gap exists in understanding how consumers actually form trust in technology-mediated traceability information. Liu, Yu and colleagues (2024) find that traceability label trust has a significant positive impact on consumers' purchasing behaviour for traceable pork products, and that food safety identification significantly strengthens this relationship, but crucially, they acknowledge that the current construction of China's pork traceability system is still at an immature stage, meaning that the consumer-behaviour findings may not yet reflect a fully functional system. Maesano, Sadrmousavigargari and Castellini (2025) further reveal that while consumers recognise blockchain's potential to improve transparency, its practical benefits are not yet fully understood or appreciated by consumers, creating a major communication and user-experience gap that technical developers have not adequately addressed.

Nguyen and Nguyen (2025) make an important theoretical contribution by developing a three-dimensional measurement scale for consumer sustainability trust, demonstrating empirically that blockchain-enabled reviews and QR code traceability systems together influence consumers' willingness to pay a premium price for sustainably traced products. However, their work simultaneously identifies that most existing research remains at the conceptual level with no validated multidimensional scale for measuring sustainability trust, representing an urgent measurement gap. Tseng *et al.*, (2022) address technology acceptance in the context of agri-food traceability systems, finding that an integrated TAM-IS success model achieves the highest explanatory power at 78.3%, yet noting that future research must replicate their China-focused findings in other national contexts and extend analysis to the manufacturer and quality-control side of the traceability equation.

The challenge of building consumer trust in food safety information is further complicated by the persistence of food fraud and counterfeiting, which undermine the credibility of even well-designed traceability systems. Doanh *et al.*, (2025) observe that the failure of most blockchain-based traceability projects is not attributable to technological immaturity but to the reluctance of supply chain partners to participate and share verified data, suggesting that stakeholder engagement and incentive design are as important as technical architecture in achieving trustworthy food traceability. Charan and Reddy (2026) highlight additional adoption barriers including technological complexity and elevated implementation costs that particularly disadvantage small-scale producers and sellers, creating unequal participation in traceability ecosystems that were theoretically designed to include all supply chain actors.

In summary, while intelligent product traceability has significant potential to enhance consumer trust and food safety, critical gaps persist in consumer technology acceptance, trust measurement, the communication of traceability benefits to non-expert audiences, the inclusion of small-scale producers, and the contextual replication of findings beyond China and high-income country settings. Future research must adopt behavioural science approaches alongside technical system design to fully bridge these gaps.

THEME 4: Intelligent Product Traceability and Consumer Trust / Food Safety

S/N	AUTHOR/DATE	TITLE	GAP STATEMENT
26	Lin et al. (2021)	<i>Consumers' intention to adopt blockchain food traceability technology towards organic food products</i>	Blockchain implementations in food supply chains lack a comprehensive roadmap. Web 3.0 and Industry 4.0 technologies enabling interoperability are still in development.
54	Nguyen & Nguyen (2025)	<i>Fostering consumer sustainability trust by integrating blockchain-enabled reviews and QR code traceability</i>	Most current research is at the conceptual level; no valid multidimensional scale exists for measuring sustainability trust in traceability contexts.
55	Charan & Reddy (2026)	<i>Enhancing Consumer Trust in Supply Chains Through Blockchain-Based Product Traceability</i>	Technological complexity and elevated implementation costs constrain adoption, especially for small-scale producers. IoT sensors for real-time monitoring need integration.
57	Liu et al. (2024)	<i>The influence of traceability label trust on consumers' traceability pork purchasing behavior</i>	China's pork traceability system is still at a low maturity level. Further validation of the effect of label-querying behaviour on continuous purchasing is needed.
58	Maesano et al. (2025)	<i>Consumer intentions to purchase organic pasta with blockchain-based traceability</i>	Blockchain's practical benefits are not yet fully understood by consumers. Future research should link traceability to directly perceived benefits like food safety.
59	Tseng et al. (2022)	<i>Understanding agri-food traceability system user intention in response to COVID-19</i>	Convenience sampling may result in bias. Research focused only on China. Quality control factors on the manufacturer's side of traceability systems need investigation.

GAP FIVE

**INTELLIGENT PRODUCT TRACEABILITY AND
ORGANISATIONAL PERFORMANCE / SUSTAINABILITY**

THEME 5:

INTELLIGENT PRODUCT TRACEABILITY

(Its Sub-Variables) and Organisational Performance / Sustainability

x1

**Operational
Efficiency**

x2

Sustainability

x3

**Counterfeit
Reduction**

x4

SC Integrity

x5

**Competitive
Advantage**

Gap 5: Intelligent Product Traceability and Organisational Performance / Sustainability

THEME 5: INTELLIGENT PRODUCT TRACEABILITY
GAP 5: INTELLIGENT PRODUCT TRACEABILITY AND ORGANISATIONAL PERFORMANCE / SUSTAINABILITY

Intelligent product traceability is increasingly recognised not only as a compliance and safety mechanism but as a strategic driver of organisational performance and long-term sustainability across multiple sectors. Khan, Parvaiz and colleagues (2022) establish empirically that supply chain transparency, enabled by traceability technology, has a positive and significant impact on organisational performance, and that traceability technology and transparency can be considered strategic resources that confer competitive advantage, an argument grounded in the Resource-Based View of the firm. This positions intelligent traceability as a value-generating capability rather than merely a regulatory cost, fundamentally reframing how organisations should invest in and manage traceability infrastructure. Onu, Mbohwa and Pradhan (2024) extend this logic by demonstrating that blockchain-based traceability can revolutionise supply chain integrity through an integrated framework emphasising visibility, authentication, and accountability, directly supporting the United Nations Sustainable Development Goal of ensuring sustainable consumption and production patterns.

Sustainability represents a growing but still insufficiently developed dimension of the traceability-performance relationship. Abobassier *et al.*, (2026) provide empirical evidence from Turkish manufacturing SMEs demonstrating that blockchain-enabled supply chain traceability positively influences sustainable supply chain practices both directly and through the mediating mechanisms of perceived information transparency and stakeholder collaboration. Their findings extend the Resource-Based View by identifying how blockchain traceability functions as a strategic resource that enables sustainability, not merely operational efficiency. Yet this remains one of the few studies to empirically test this relationship, and its geographical and sectoral scope is narrow. Adedugba *et al.*, (2022) similarly identify that inventory-related traceability practices are significantly linked to environmental sustainability through carbon emission reduction, but call for coordinated multi-echelon traceability frameworks to fully realise sustainability potential across the supply chain.

Counterfeit reduction is one of the most direct and measurable organisational benefits of intelligent product traceability, yet the research evidence remains fragmented and largely limited to case studies and theoretical frameworks. Onu *et al.*, (2024) document that the infiltration of counterfeit products into global supply chains spans pharmaceuticals, consumer electronics, and luxury goods, undermining brand reputation, consumer trust, and public safety, yet the empirical evidence base for blockchain-based anti-counterfeiting traceability in real-world, large-scale environments is still thin. Biglia *et al.*, (2022) demonstrate the application of neural networks coupled with traceability systems to detect product substitution fraud in the meat industry with up to 100% accuracy in laboratory settings, but acknowledge that the methodology requires further robustness improvements and integration with blockchain systems before commercial deployment. The laboratory-to-field translation gap is pervasive across performance-focused traceability research.

Ali, Shin and Song (2022) address organisational performance in the context of smart manufacturing, demonstrating that a blockchain-enabled open quality system achieves high transaction throughput and transparency in quality data management, yet their evaluation was conducted in a simulated environment, leaving economic implications including return on investment and cost of implementation entirely unanalysed. This reflects a broader methodological gap across the organisational performance literature on traceability: studies predominantly focus on technical performance metrics such as throughput and latency rather than financial, competitive, or strategic performance outcomes. Harkonen, Guerrero Rodriguez and Mustonen (2024) identify a related conceptual gap, noting that the product management perspective on end-to-end traceability, specifically the integration of technical and operational traceability across the full product lifecycle, has not been extensively studied. While intelligent product traceability contributes to organisational performance and sustainability across multiple dimensions, significant gaps persist in the empirical evidence base for financial and strategic performance outcomes, the sustainability impact of traceability practices beyond a narrow set of industries, the commercial scalability of laboratory-validated counterfeit-detection systems, and the longitudinal assessment of traceability's contribution to competitive advantage in emerging market contexts including Nigeria. Future research must extend beyond technical performance benchmarks to address the broader strategic and sustainability dimensions of intelligent product traceability.

THEME 5: Intelligent Product Traceability and Organisational Performance / Sustainability

S/N	AUTHOR/DATE	TITLE	GAP STATEMENT
19	Harkonen et al. (2024)	<i>The role of productization in end-to-end traceability</i>	Notable lack of focus on the integration of technical and operational traceability across the full product lifecycle.
30	Biglia et al. (2022)	<i>Artificial intelligence to boost traceability systems for fraud prevention in the meat industry</i>	Methodology can be further improved in robustness to fraud attempts. Integration with blockchain to guarantee inviolability of traceability data is needed.
40	Onu et al. (2024)	<i>Blockchain-powered traceability solutions: pioneering transparency to eradicate counterfeit products</i>	Small sample size of interview participants. The framework has not been empirically tested in large-scale real-world environments across global industries.
41	Ali et al. (2022)	<i>Blockchain-enabled open quality system for smart manufacturing: applications and challenges</i>	Evaluation conducted in simulated environment. Economic implications including cost of implementation and return on investment require further investigation.
44	Abobassier et al. (2026)	<i>Unlocking Sustainable Supply Chains Through Blockchain Traceability</i>	Cross-sectional data limits causal inference. Findings focused on Turkish manufacturing SMEs and need verification in large corporations and other regions.
56	Doanh et al. (2025)	<i>Blockchain-based food traceability system and green brand image</i>	Findings based on literature analysis only; empirical research and real-world case studies are needed. Scope is limited to the manufacturing industry.